

Hao-Ran Liu | Full-Stack Engineer

National Taichung University of Science and Technology | Computer Science and Information Engineering Master

Hsinchu City | 6.2 years of work experience | 0908-226-952 | kn880701@gmail.com

Profile

I am Hao-Ran Liu, a full-stack engineer familiar with iOS / Android App development, Web frontend, backend services, and automated deployment process construction.

In projects, I am not only responsible for feature development but also frequently participate in architecture planning, cross-platform integration, and post-launch maintenance. With experience in App, frontend, backend, and deployment, I can understand requirements from a complete system perspective, plan data flows, and implement stable and scalable product features.

In practical development, I utilize AI Agents to assist and improve efficiency; meanwhile, I continue to refine my system design, code architecture, and problem analysis skills to ensure I can assess the feasibility and security risks of AI-generated code, correcting direction promptly when it deviates from requirements or introduces potential vulnerabilities.

I believe my advantage lies in quickly understanding product requirements and planning stable, maintainable, and scalable technical solutions from a full-stack perspective. I expect to continue accumulating technical depth and creating practical value in challenging product and system development.

Specialties

Mobile Development

Cross-platform mobile application development. Specialized in building Android and iOS applications.

iOS (Flutter) | Android (Kotlin / Java / Flutter) | Jetpack | Android Unit Test | Appium (Automated Testing)

Backend Technologies

Proficient in Go, with practical experience in PostgreSQL, Docker containerization, CI/CD pipeline construction, and AI service integration.

Go (Gin/Gorm) | Flask | RESTful API Design | JWT Authentication

Web Development

Developing web interfaces using React, and LINE LIFF development.

React | LINE LIFF

Architecture Design

Familiar with MVVM, MVP, and Clean Architecture patterns. Capable of planning highly cohesive and loosely coupled system structures for various project sizes to effectively separate business logic from UI.

MVVM | MVP | Clean Architecture

CI/CD & Deployment

Experienced in automated deployment and cloud service integration. Able to use Docker Compose for containerized environment setup and build CI/CD pipelines for automated testing and deployment.

Docker Compose GitHub Actions Automation GCP

Honors & Awards



2026 MediaTek 'Genius for Home' Potential Pioneer Award

Winning Project: 母語也會通 (Taiwanese Learning Platform)

Awarded by MediaTek Foundation for innovative use of AI speech recognition and gamification in indigenous language digitalization.



2022 MediaTek 'Genius for Home' 1st Prize

Winning Project: Dental Guard

Won 1st Prize in Taiwan's premier social tech competition for utilizing AI oral plaque detection to enhance healthcare accessibility.

Work Experience

Total Experience: 6.2 years of work experience

Full-Stack Engineer | Acer Synergy Tech

2025/10 - Present

- Served as a full-stack engineer, responsible for maintaining and developing mobile (Flutter), frontend (React), and backend (Go) systems.
- Developed the LINE LIFF course registration system, replacing manual inquiry and registration processes.
- Developed the official App for Hsinchu Lioneers, successfully serving thousands of active users and ensuring a smooth cross-platform experience.
- Responsible for service containerization and cloud environment deployment, building automated update processes to ensure the convenience of system maintenance.

APP Engineer | Ajie Technology

2025/03 - 2025/09

- Developed a customized Android browser based on the Chromium open-source project.
- Wrote Android Unit Tests using Java, achieving over 50% test coverage to ensure code quality and stability.

APP Engineer | Clockwork Orange Technology Co., Ltd.

2023/11 - 2025/02

- Served as the primary developer of the APP team, delivering over 5 Kotlin/Java and Flutter applications.
- Led the development of IoT smart device Apps, handling complex BLE / BluFi Bluetooth communication protocols, optimizing the disconnection reconnection mechanism, and increasing the hardware connection success rate to over 90%.
- Conducted technology selection and refactoring while ensuring system fluency, completing cross-platform programming, reducing development and maintenance costs for the team by over 50%.

Intern Engineer | National Taichung University of Science and Technology

2020/09 - 2023/07

- Participated in industry-academia cooperation as an intern engineer, developed and maintained campus systems and Android Apps using Java and Kotlin, covering areas such as IoT data streaming, Bluetooth connectivity, healthcare, and robot control.
- Participated in the FDA collaboration project Mobile FDA: Used Flutter to assist in digitizing the traditional paper inspection process, introduced offline data storage and electronic signature functions for vendors, significantly improving the government's inspection administrative efficiency.
- Participated in campus collaboration projects such as the "Delta Robot Assistant", supporting system design and feature development.
- Participated in projects like "Dr. Voice Education" and "Dental Guard", responsible for mobile application development and device integration.
- Collaborated with the National Police Agency on a project, using Kotlin to develop an App to intercept and analyze network packets and UI behavior.

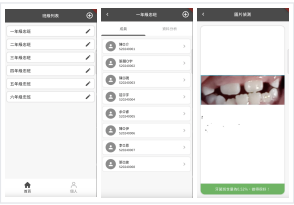
Project Achievements



母語也會通 (Mó-gí lā-ē-thong)

Won the Potential Pioneer Award at MediaTek's 'Genius for Home' 2026. A gamified Taiwanese language learning platform featuring pronunciation practice with AI voice recognition, Tâi-lô romanization training, and real-time friend battle mode.

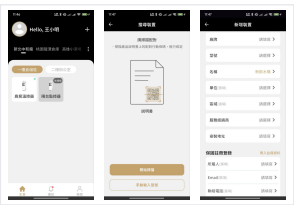
- Won the Potential Pioneer Award at 2026 MediaTek 'Genius for Home' competition
- Promoted Taiwanese cultural heritage through gamified design
- Integrated voice recognition and Tâi-lô system for immersive learning



Dental Guard

Won the 1st prize in MediaTek's 'Genius for Home' in 2022. Integrated Flutter APP, AI dental plaque image recognition, and Go backend service, introducing Docker and CI/CD workflows.

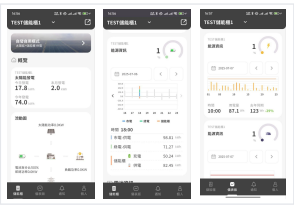
- Won the first prize in the 2022 MediaTek "Genius for Home" competition
- Successfully integrated the AI model and mobile application
- Established a complete DevOps process



Smart Kitchen Appliances

An IoT application developed with Flutter, connecting BluFi hardware communication and a Go backend to achieve smart kitchen appliance control, successfully launched on dual platforms.

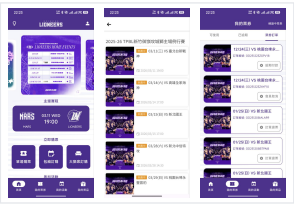
- Successfully launched on iOS and Android platforms
- Stable IoT device communication mechanism
- Excellent user experience design
- Successful commercial product case



EMS Energy Management System

Responsible for overall system architecture planning, developed customized dynamic energy charts with Flutter, and integrated third-party logins and energy storage cabinet hardware.

- Complete system architecture planning
- Successfully integrated multiple third-party services
- Provided intuitive data visualization



Lioneers AST Team Service App

A one-stop mobile service portal for Hsinchu Lioneers fans, integrating memberships, events, tickets, and team-related info.

- Established a unified mobile service portal for the team and fans
- Integrated push notifications and Deep Links to boost event and match promotion
- Built a sustainable and extensible modular Flutter project architecture

Education

National Taichung University of Science and Technology
2021/09 ~ 2023/07

Computer Science and Information Engineering | Master

- Won the first prize in the MediaTek "Genius for Home" Competition in 2022 by developing the "Dental Guard".
- Focused on AI image recognition and deep learning application research, and published 3 related academic papers.

National Taichung University of Science and Technology
2017/09 ~ 2021/06

Computer Science and Information Engineering | Bachelor

- Worked as an intern engineer during studies, accumulating practical development experience in Android App (Java/Kotlin) and IoT system integration.

著作與論文發表

Human movement science-informed multi-task spatio temporal graph convolutional networks for fitness action recognition and evaluation

Applied Soft Computing (Elsevier) • 2024

Extending the core algorithm of my master's thesis, it was successfully published in the authoritative international SCI journal "Applied Soft Computing" (Elsevier, 2024). This study proposed a multi-task Spatio-Temporal Graph Convolutional Network (ST-GCN) architecture incorporating human kinematics, which can not only accurately recognize fitness action classification, but also introduce a multi-task objective function for quantitative evaluation and guidance of action quality, proving high robustness and accuracy on real-world datasets collected in gyms.

Published in SCI International Journal | Multi-task Learning | Action Quality Assessment and Guidance System

Using 5PKC-based Skeleton Partition Strategy in Spatio-Temporal Graph Convolution Networks for Optimizing Fitness Action Recognition and Evaluation

Master's Program, Dept. of CSIE, National Taichung University of Science and Technology • 2023

Master's thesis for graduation from National Taichung University of Science and Technology. Aiming at the defect that traditional Convolutional Neural Network (CNN) image analysis is easily interfered by light and background, this study innovatively established the "Five Principal Kinematic Chains (5PKC)" skeleton partition algorithm. Through the analysis of the spatiotemporal correlation of the 3D skeleton by the Spatio-Temporal Graph Convolutional Network (ST-GCN), a high-precision home fitness action recognition system prototype that can eliminate background noise was successfully designed.

Spatio-Temporal Graph Convolutional Network (ST-GCN) | 5PKC Skeleton Partition Strategy | Human Action Recognition

Applying 5PKC-Based Skeleton Partition Strategy into Spatio-Temporal Graph Convolution Networks for Fitness Action Recognition

Springer (Innovative Computing Vol 2 - Emerging Topics in Future Internet / IC 2023) • 2023

This study focuses on applying the Five Principal Kinematic Chains (5PKC) skeleton partition strategy to Spatio-Temporal Graph Convolutional Networks (ST-GCN), focusing on optimizing home fitness action recognition, solving the problem of traditional image processing being easily interfered by background noise, and conducting experimental comparisons with other cutting-edge methods.

Included in Springer's international academic book series | Published at an international academic conference (IC 2023) | Intelligent Computing and Computer Vision

Certifications

Level B technician for computer software application (Workforce Development Agency, MOL)
ANDES Embedded System Developer Certification (Andes Technology)
Professional Vocabulary Quotient Credential (PVQC) (GLAD)
IPAS IoT Application Engineer - Primary (Ministry of Economic Affairs)